



ASSIGNMENT 2

Loan Approval Prediction Using Classification

OBJECTIVE

Build, train, evaluate and test a classification model that predicts whether a loan application is likely to be Approved or Rejected.

PROJECT OVERVIEW

Business Problem

A bank receives loan applications from customers. Use historical application data to predict whether a new application will be Approved or Rejected.

INPUT FEATURES	TARGET
Gender	Loan Status
Married	Y = Approved
Dependents	N = Rejected
Education	
Self Employed	
Applicant Income	
Coapplicant Income	
Loan Amount / Loan Term	
Credit History / Property Area	

RECOMMENDED ALGORITHM

Primary Algorithm

Logistic Regression — implement this as the main classification model. Optionally compare with Decision Tree or Random Forest.

TECHNOLOGY STACK

Python	NumPy	Pandas
Matplotlib	Seaborn	Scikit-learn

Project Flow

Loan Data → Preprocessing → Classification Model → Approved / Rejected

ASSIGNMENT TASKS

Complete all tasks. Submit clean code, outputs, visualizations, evaluation results and a short conclusion.

01	Understand the Business Problem	Define the loan-approval problem and prediction target.
02	Collect / Download Dataset	Download a suitable Loan Approval dataset.
03	Load Dataset Using Pandas	Read the dataset and inspect the first records.



04	Perform Data Understanding	Use head(), shape, info(), describe() and missing-value checks.
05	Perform EDA	Explore income, loan amount, credit history, education, property area and target distribution.
06	Identify X and y	Select independent variables X and target y = Loan_Status.
07	Handle Missing Values	Identify and handle missing values using suitable techniques.
08	Remove Unnecessary Columns	Remove identifiers or irrelevant columns.
09	Encode Categorical Data	Convert Gender, Education, Property Area and other categorical features into numerical form.
10	Train / Test Split	Split data into 80% training and 20% testing.
11	Build Logistic Regression	Create a Scikit-learn LogisticRegression() model.
12	Train the Model	Train the model using the training dataset.
13	Make Predictions	Predict Approved or Rejected for test applications.
14	Evaluate the Model	Calculate Accuracy, Precision, Recall, F1-Score and create a Confusion Matrix.
15	Visualize Results	Create target-distribution, feature-relationship and confusion-matrix visualizations.
16	Test New Application	Create a new applicant record and predict Approved or Rejected.
17	Write the Conclusion	Summarize performance, observations, limitations and improvements.

TECHNICAL REQUIREMENTS

EDA & DATA UNDERSTANDING

- head() — first records.
- shape — rows and columns.
- info() — columns and data types.
- describe() — numerical summary.
- Missing values — identify and handle nulls.
- Target distribution — Approved vs Rejected counts.

Analysis	Recommended Visualization
Loan Status	Count Plot / Bar Chart
Applicant Income	Histogram
Loan Amount	Histogram / Box Plot
Credit History vs Status	Count Plot
Numerical relationships	Correlation Heatmap

CLASSIFICATION METRICS

Metric	Meaning	Better
Accuracy	Overall correct predictions.	Higher



Precision	Correct positive predictions among predicted positives.	Higher
Recall	Correctly identified positive cases.	Higher
F1-Score	Balance between Precision and Recall.	Higher
Confusion Matrix	Correct and incorrect classification counts.	Interpret

NEW LOAN APPLICATION

Feature	Sample Value
Gender	Male
Married	Yes
Dependents	0
Education	Graduate
Applicant Income	₹50,000
Loan Amount	₹2,00,000
Credit History	1
Property Area	Urban

Prediction Flow

New Applicant → Trained Model → Loan Approved / Rejected

Important

Educational project only. Real lending systems require fairness, explainability, regulatory compliance and human oversight.

SUBMISSION & LEARNING OUTCOMES

Expected Learning Outcomes

01	Understand a real-world Classification problem.
02	Understand how ML can support loan-approval decisions.
03	Load and preprocess data using Pandas.
04	Handle missing values and categorical features.
05	Perform EDA and identify patterns.
06	Separate X and y.
07	Build and train Logistic Regression.
08	Make predictions for unseen applications.
09	Evaluate Accuracy, Precision, Recall and F1-Score.
10	Interpret a Confusion Matrix.
11	Test a new loan application.
12	Explain model performance and limitations.

Submission Checklist

01	Dataset collected	STATUS
02	Dataset loaded	STATUS
03	Data understanding completed	STATUS



04	Missing values handled	STATUS
05	EDA completed	STATUS
06	X and y identified	STATUS
07	Unnecessary columns removed	STATUS
08	Categorical features encoded	STATUS
09	80/20 split completed	STATUS
10	Logistic Regression created	STATUS
11	Model trained	STATUS
12	Predictions generated	STATUS
13	Accuracy calculated	STATUS
14	Precision calculated	STATUS
15	Recall calculated	STATUS
16	F1-Score calculated	STATUS
17	Confusion Matrix interpreted	STATUS
18	New application tested	STATUS
19	Conclusion written	STATUS

STUDENT CHALLENGE

Be ready to explain: Why is this a Classification problem? Why is Logistic Regression suitable? What does Credit History contribute?

Goal: Explain your project from Business Problem → Data → Preprocessing → Model → Prediction → Evaluation.

Assignment Standard: Submit clean code, meaningful visualizations, evaluation metrics, observations and a clear conclusion.